Class	$[0;10[$	$[10;20[$	$[20;30[$
Count	4	9	7

What are the total frequency, the modal class and the frequency of $[10;20[$?

Ans. 20 ; $[10;20[$; $\dfrac{9}{20} = 45\,\%$.

5. Running thread · Problem [DIV] One bus comes every 15 min, another every 20 min. They meet at 8:00. What time do they meet again?

Ans. LCM $(15\,;\,20) = 60$: at9 h.

Sheet 17 Mon. 30 Nov. 2026 Absolute value and distance between two real numbers

1. Warm-up · Flash [REP] What is the distance between -3 and 5 on the number line?

Ans. $|5-(-3)| = 8$.

2. Core · Flash [CAL] Calculate $|-7|$, $|3-8|$ and $|\pi-4|$.

Ans. $7\,;\,5\,;\,|\pi-4| = 4-\pi$ since $\pi < 4$.

3. Core + · Method [EQ] Solve $|x-3| = 2$.

Ans. The real numbers at distance 2 from 3: $x = 1$ or $x = 5$.

Geometrical interpretation rather than case by case.

4. Review · Formula [IMG] $f(x) = x^2 - 3$. Give the image of -2 and the preimages of 6.

Ans. $f(-2) = 1$; $x^2 - 3 = 6 \Leftrightarrow x^2 = 9$: preimages -3 and 3.

5. Running thread · Formula → table [REG] $f(x) = -x + 3$. Complete: $f(-1), f(0), f(2)$.

Ans. $4\,;\,3\,;\,1$.

Sheet 18 Tue. 1 st Dec. 2026 Inequality $|x-a| \leqslant$ r and associated interval

1. Warm-up · Flash [EQ] Solve $2x - 3 \leqslant 5$.

Ans. $2x \leqslant 8$, $x \leqslant 4$: $S =]-\infty;4]$.

2. Core · Method [EQ] Solve $|x-2| \leqslant 3$ and give S as an interval.

Ans. The real numbers at distance at most 3 from 2: $S = [-1;5]$.

Check: $|-1-2| = 3$ and $|5-2| = 3$.

3. Core + · Translate [NUM] Express as an inequality with an absolute value: $x \in [-1;5]$ then $x \in]4;10[$.

Ans. $|x-2| \leqslant 3$ (centre 2, radius 3); $|x-7| < 3$ (centre 7, radius 3).

4. Review · Problem [CMP] Pack A: 6 yoghurts for 4.20 €. Pack B: 10 yoghurts for 6.80 €. Which is the better deal?

Ans. A : 0,70 € each; B: 0.68 € : B.

5. Running thread · Flash [GEO] A disc has radius 5 cm. Give its area and its perimeter (exact values).

Ans. Area $= \pi \times 5^2 = 25\pi$ cm^2 ; perimeter $= 2\pi \times 5 = 10\pi$ cm.

Functions: representations and reference functions

Sheet 19 Thu. 3 Dec. 2026 Real-valued function; domain

1. Warm-up · Flash [EQ] Solve $x - 4 = 0$, then 2 $x + 6 = 0$.

Ans. $x = 4$; $x = -3$.

2. Core · Method [MOD] Domain of $f(x) = \dfrac{1}{x-4}$?

Ans. We exclude the value at which the denominator is zero: $\mathbb{R} \setminus \{4\} =]-\infty;4[\cup]4;+\infty[$.

3. Core + · MCQ [MOD] The domain of definition of $h(x) = \dfrac{1}{x^2+1}$ is:

A. $\mathbb{R} \setminus \{1\}$ **B.** $\mathbb{R} \setminus \{-1\,;\,1\}$ **C.** $\mathbb{R}$ **D.** $\mathbb{R}^*$

Ans. C. $\mathbb{R}$

$x^2 + 1 \geqslant 1 > 0$: the denominator never vanishes.

4. Review · Table [PNP] Is this a table of proportional values?

x	2	5	8
y	6	15	24

Ans. Yes: $\frac{y}{x} = 3$ for each column.

5. Running thread · Reading code [ALG] Output of $f(4)$?

```
def f(x):
    return 3*x - 2
```

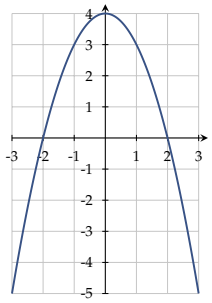
Ans. 10.

Sheet 20 Mon. 7 Dec. 2026 Graph with equation y $= f(x)$; images and preimages

1. Warm-up · Flash [IMG] $f(x) = -x^2 + 4$. Calculate $f(0)$, $f(-2)$ and $f(3)$.

Ans. 4 ; 0 ; -5.

2. Core · Graph [IMG] Graph of f :



Read $f(1)$ and the preimages of 0.

Ans. $f(1) = 3$; preimages of 0: -2 and 2.

Expression used for the plot: f $(x) = -x^2 + 4$.

3. Core + · Reflection [IMG] For this function, does the equation $f(x) = 5$ have solutions?

Ans. No: $f(x) = 4 - x^2 \leqslant 4 < 5$ for all x ; the curve does not rise as far as $y = 5$.

4. Review · Flash [PRB] Urn: 5 red balls, 3 blue, 2 green. Probability of drawing a blue one? A non-red?

Ans. $\frac{3}{10}$; $\frac{5}{10} = \frac{1}{2}$.

5. Running thread · Method [EQ] Solve in $\mathbb{R}$: $4 - 3x = 10$.

Ans. $x = -2$

Pitfall: dividing by -3 without changing the sign of the result.

Sheet 21 Tue. 8 Dec. 2026 Reference functions: absolute value, square, reciprocal

1. Warm-up · Flash [CAL] Calculate $|-2{,}5|$, $(-3)^2$ and $\frac{1}{4}$.

Ans. $2{,}5$; 9 ; $-0{,}25$.

2. Core · Table [REG] Complete for $x = -2$; -1; 0; 1; 2 the values of $x \mapsto |x|$, $x \mapsto x^2$ & $x \mapsto \frac{1}{x}$.

Ans. $|x|$: 2 ; 1 ; 0 ; 1 ; 2 ; x^2 : 4 ; 1 ; 0 ; 1 ; 4 ; $\frac{1}{x}$: $-\frac{1}{2}$; -1; undefined; 1; 1 $\frac{}{2}$.

3. Core + · MCQ [REG] Which reference curve passes through $(-1;1)$, $(0;0)$, $(1;1)$ and $(2;4)$?

A. the line $y = x$ **B.** that of the reciprocal **C.** that of the absolute value **D.** that of the square

Ans. D. that of the square

The absolute value would pass through $(2;2)$; the reciprocal is not defined at 0 ; the line y $= x$ passes through $(-1;-1)$.

4. Review · Flash [PRO] An object is enlarged by scale factor 1.5. What does a length of 8 cm become? By what number is the area multiplied by?

Ans. 12 cm; the area is multiplied by $1.5^2 = 2{,}25$.

5. Running thread · Prove [LIT] Conjecture: "the sum of two consecutive integers is odd". Prove it.

Ans. $n + (n+1) = 2n + 1$, of the form $2k + 1$ with $k = n$: odd.

Sheet 22 Thu. 10 Dec. 2026 Square root and cube; sign of the reference functions

1. Warm-up · Flash [CAL] Calculate $\overline{49}$, $\sqrt{0.25}$ and $(-2)^3$.

Ans. 7 ; $0{,}5$; -8.

2. Core · Method [MOD] Study the sign of $f(x) = -2x + 6$.

Ans. $-2x + 6 = 0 \Leftrightarrow x = 3$; $f(x) > 0$ for $x < 3$, $f(x) < 0$ for $x > 3$ (negative slope).

3. Core + · True/false [MOD] True or false? Justify. For every real number x, $\sqrt{x^2} = x$.

Ans. False. Counterexample: $x = -3$: $\sqrt{9} = 3 \neq -3$. In fact $\sqrt{x^2} = |x|$.

4. Review · Flash [PER] How many squares make up a net of a cube? Does a cross (row of 4 squares + 1 square on each side) work?

Ans. 6 squares; yes, the cross folds into a cube.

5. Running thread · Flash [SOL] How many faces, vertices and edges does a square-based pyramid have?

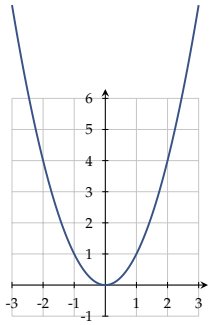
Ans. 5 faces, 5 vertices, 8 edges.

Sheet 23 Mon. 14 Dec. 2026 Graphical and numerical solving: f $(x) = k$, $f(x) < k$

1. Warm-up · Flash [EQ] Solve $x^2 = 16$ then $x^2 = -4$.

Ans. $x = \pm 4$; no real solution ($x^2 \geqslant 0$).

2. Core · Graph [REG] Graph of $f(x) = x^2$:



Solve graphically $f(x) = 4$ then $f(x) < 4$.

Ans. $f(x) = 4 : x = -2$ or $x = 2$; $f(x) < 4 : x \in]-2;2[$.

Find the x-coordinates of the points of the curve below the line y $= 4$.

3. Core + · Method [EQ] Solve $|x| < 3$ then $|x| = 5$.

Ans. $]-3;3[$; $x = -5$ or $x = 5$.

4. Review · Pitfall [SUB] Calculate $-x^2$ then $(-x)^2$ for $x = -3$.

Ans. $-(-3)^2 = -9$ and $(-(-3))^2 = 9$.

5. Running thread · Method [DEM] In a triangle ABC, M and N are the midpoints of $[AB]$ and $[AC]$. What do we prove? Which property do we quote?

Ans. $(MN) \parallel (BC)$ and $MN = \frac{1}{2}BC$: midpoint theorem.